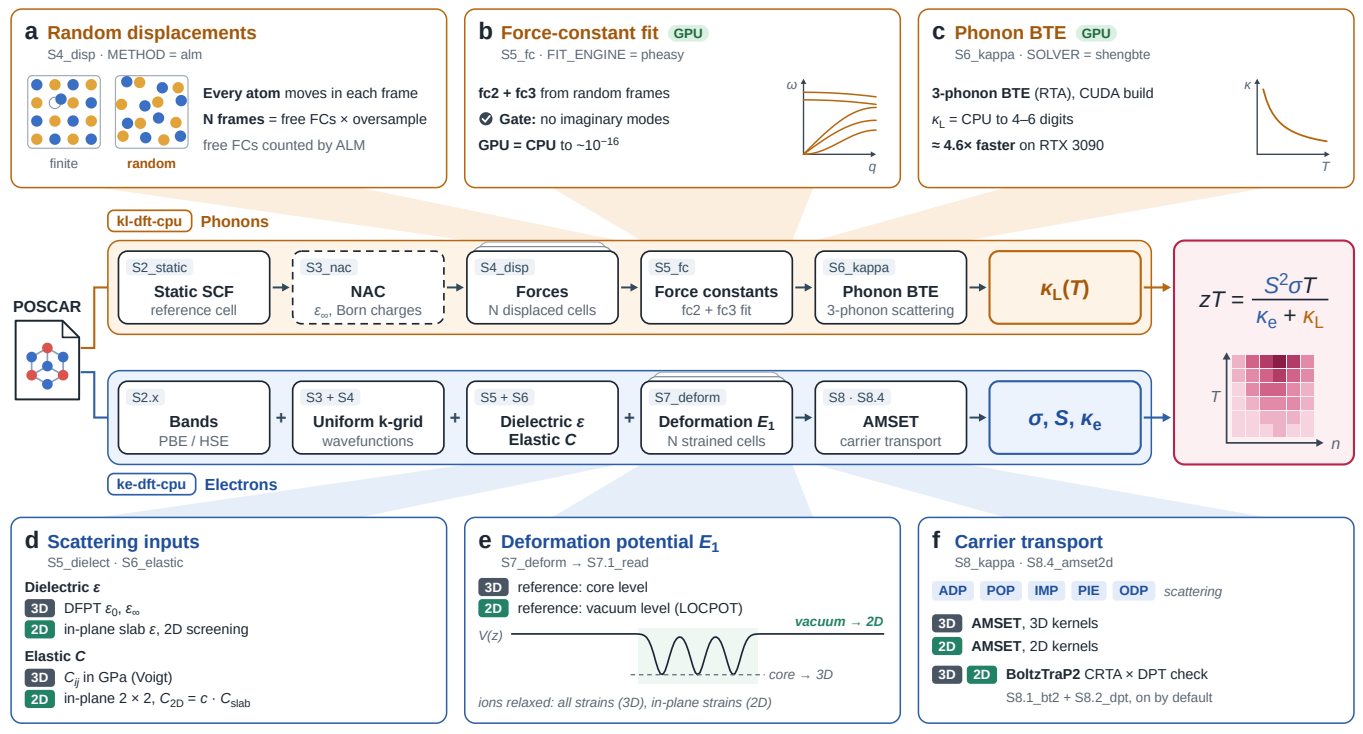

a Random displacements
S4_disp · METHOD = alm
finite
random
Every atom moves in each frame
N frames = free FCs × oversample
free FCs counted by ALM
b Force-constant fit GPU
S5_fc · FIT_ENGINE = pheasy
fc2 + fc3 from random frames
Gate: no imaginary modes
GPU = CPU to ~10−16
ω
q
c Phonon BTE GPU
S6_kappa · SOLVER = shengbte
3-phonon BTE (RTA), CUDA build
κL = CPU to 4–6 digits
≈ 4.6× faster on RTX 3090
κ
T
kl-dft-cpu Phonons
POSCAR
S2_static
Static SCF
reference cell
S3_nac
NAC
ε∞, Born charges
S4_disp
Forces
N displaced cells
S5_fc
Force constants
fc2 + fc3 fit
S6_kappa
Phonon BTE
3-phonon scattering
κL(T)
zT = S²σT / (κe + κL)
T
n
S2.x
Bands
PBE / HSE
S3 + S4
Uniform k-grid
wavefunctions
S5 + S6
Dielectric ε
Elastic C
S7_deform
Deformation E1
N strained cells
S8 · S8.4
AMSET
carrier transport
σ, S, κe
ke-dft-cpu Electrons
d Scattering inputs
S5_dielect · S6_elastic
Dielectric ε
3D DFPT ε0, ε∞
2D in-plane slab ε, 2D screening
Elastic C
3D Cij in GPa (Voigt)
2D in-plane 2 × 2, C2D = c · Cslab
e Deformation potential E1
S7_deform → S7.1_read
3D reference: core level
2D reference: vacuum level (LOCPOT)
V(z)
vacuum → 2D
core → 3D
ions relaxed: all strains (3D), in-plane strains (2D)
f Carrier transport
S8_kappa · S8.4_amset2d
ADP POP IMP PIE ODP scattering
3D AMSET, 3D kernels
2D AMSET, 2D kernels
3D 2D BoltzTraP2 CRTA × DPT check
S8.1_bt2 + S8.2_dpt, on by default